Inference

Raffaele Saggio

UBC

October 14, 2026

Section 1

Statistics Boot Camp

Statistical Properties of OLS

- We've talked about the population OLS projection as an approximation to the CEF
- This didn't require any assumptions about the data-generating process
- Next, we'll talk about the statistical properties of the OLS **estimator** based on a sample of data, and how these depend on the DGP

The OLS Estimator

- Let's again work with the projection model:

$$Y_i = X_i'\beta + U_i,\ E\left[X_i U_i\right] = 0$$

- We have a sample of N individuals from the population
- The OLS estimator of β is

$$\hat{\beta} = \left(\frac{1}{N}\sum_i X_i X_i'\right)^{-1}\left(\frac{1}{N}\sum_i X_i Y_i\right)$$

$$= \left(\frac{1}{N}\sum_i X_i X_i'\right)^{-1}\left(\frac{1}{N}\sum_i X_i (X_i'\beta + U_i)\right)$$

$$= \beta + \left(\frac{1}{N}\sum_i X_i X_i'\right)^{-1}\left(\frac{1}{N}\sum_i X_i U_i\right)$$

OLS in Finite Samples

- First consider the finite-sample properties of $\hat{\beta}$
- Its expectation is

$$E\left[\hat{\beta}\right] = \beta + E\left[\left(\frac{1}{N}\sum_i X_i X_i'\right)^{-1}\left(\frac{1}{N}\sum_i X_i U_i\right)\right]$$

$$\neq \beta + \left(\frac{1}{N}\sum_i E\left[X_i X_i'\right]\right)^{-1}\left(\frac{1}{N}\sum_i E\left[X_i U_i\right]\right)$$

$$= \beta$$

- In general $\hat{\beta}$ is a biased estimate of the population projection coefficient vector β
- An exception is when the CEF is linear so $E\left[Y_i|X_i\right] = X_i'\beta$ (just apply the law of iterated expectations)

Fixed vs. Stochastic Regressors

- You may be surprised to learn that OLS is biased in finite samples for the population projection
- The properties of OLS are traditionally derived assuming that the regressor matrix X is non-stochastic, i.e., fixed in repeated samples
- Sometimes (e.g. in an experiment) the researcher gets to control the distribution of the regressors
- When X is non-stochastic we have

$$E\left[\hat{\beta}\right] = \beta + E\left[\left(\frac{1}{N}\sum_i X_i X_i'\right)^{-1}\left(\frac{1}{N}\sum_i X_i U_i\right)\right]$$

$$= \beta + \left(\frac{1}{N}\sum_i X_i X_i'\right)^{-1}\left(\frac{1}{N}\sum_i E\left[X_i U_i\right]\right)$$

$$= \beta$$

Fixed vs. Stochastic Regressors

- Whether it makes sense to consider the regressors fixed or random depends on the context
- Do we want a best approximation to the CEF using the distribution of X_i in our sample, or in the population?
  - Note: sometimes your sample is "the population" (e.g. everyone in Iowa)
- For now let's consider the traditional fixed-regressor case

The Variance of OLS

- In matrix form the variance of the OLS estimator is

$$E\left[\left(\hat{\beta}-E\left[\hat{\beta}\right]\right)\left(\hat{\beta}-E\left[\hat{\beta}\right]\right)'\right]=E\left[\left(\hat{\beta}-\beta\right)\left(\hat{\beta}-\beta\right)'\right]$$

$$=E\left[\left(X'X\right)^{-1}X'UU'X\left(X'X\right)^{-1}\right]$$

$$=\left(X'X\right)^{-1}X'E\left[UU'\right]X\left(X'X\right)^{-1}$$

$$=\left(X'X\right)^{-1}X'\Omega X\left(X'X\right)^{-1}$$

- Ω is the $N\times N$ symmetric covariance matrix with ij-th element $Cov(U_i,U_j)$

The Variance of OLS

- The "meat" in the middle of our sandwich covariance matrix can be written

$$X'\Omega X = \sum_i X_i X_i' \sigma_i^2 + 2 \sum_{i<j} X_i X_j' \sigma_{ij}$$

- Here $\sigma_i^2 = Var(U_i)$ and $\sigma_{ij} = Cov(U_i, U_j)$
- There are $N(N+1)/2$ unknown terms in this expression
- Estimating all of these terms is intractable without further assumptions

The Variance of OLS

- Suppose the regression errors are independent and homoskedastic, so $Cov\left(U_i, U_j\right) = 0$ and $Var\left(U_i\right) = \sigma^2 \ \forall i$. Then

$$Var\left(\hat{\beta}\right) = \left(X'X\right)^{-1} X' \left(\sigma^2 I\right) X \left(X'X\right)^{-1}$$

$$= \sigma^2 \left(X'X\right)^{-1}$$

- The conventional estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{1}{N-K} \sum_i \hat{U}_i^2$$

Degrees of Freedom Correction

$$\hat{\sigma}^2 = \frac{1}{N-K} \sum_i \hat{U}_i^2$$

- Why $1/(N-K)$ instead of $1/N$?
- This is a degrees of freedom correction to account for finite-sample bias
- $\hat{U} = M_X Y$, so

$$\begin{aligned} Var\left(\hat{U}\right) &= E\left[M_X Y Y' M_X\right] \\ &= M_X E\left[UU'\right] M_X \\ &= \sigma^2 M_X. \end{aligned}$$

- Then $Var\left(\hat{U}_i\right) = \sigma^2(1 - P_{ii})$, where P_{ii} is the i'th diagonal entry in P_X

Degrees of Freedom Correction

$$Var\left(\hat{U}_i\right) = \sigma^2(1 - P_{ii})$$

- It turns out that $\sum_i P_{ii} = K$:

$$\sum_i P_{ii} = tr\left(X\left(X'X\right)^{-1}X'\right) = tr\left(X'X\left(X'X\right)^{-1}\right) = tr(I) = K$$

- This means that the expectation of the mean squared residual is

$$E\left[\frac{1}{N}\sum_i \hat{U}_i^2\right] = \frac{1}{N}\sum_i E\left[\hat{U}_i^2\right] = \frac{1}{N}\sum \sigma^2(1 - P_{ii}) = \sigma^2 \cdot \frac{N-K}{N}$$

- We therefore need to multiply by $(N/(N-K))$ to get an unbiased estimate of σ^2 (just like we divide by $N-1$ instead of N to get unbiased variance estimates)

Finite-sample Inference

- We've shown how to get an unbiased estimate of $\hat{\sigma}^2$, and therefore of $Var\left(\hat{\beta}\right)$, under homoskedasticity
- But this isn't enough to allow us to test hypotheses or form confidence intervals
- To conduct inference we need to know the distribution of $\hat{\beta}$
- The traditional approach is to assume that the U_i are *iid* and normally distributed, which implies that $\hat{\beta}$ is normal and $\left(\hat{\beta}_k - \beta_k\right) / \hat{Var}\left(\hat{\beta}_k\right)$ follows a t-distribution with $(N - K)$ degrees of freedom
- Inference without parametric assumptions (usually) requires an asymptotic approximation

OLS Asymptotics

- Let's again express the OLS estimator as

$$\hat{\beta} = \beta + \left(\frac{1}{N}\sum_i X_i X_i'\right)^{-1}\left(\frac{1}{N}\sum_i X_i U_i\right)$$

- This can be rewritten

$$\sqrt{N}\left(\hat{\beta} - \beta\right) = \left(\frac{1}{N}\sum_i X_i X_i'\right)^{-1}\left(\frac{1}{\sqrt{N}}\sum_i X_i U_i\right)$$

- As $N \to \infty$ this quantity has the same distribution as:

$$E\left[X_i X_i'\right]^{-1}\left(\frac{1}{\sqrt{N}}\sum_i X_i U_i\right)$$

- Note: all asymptotic variability through the "score" $\sum_i X_i U_i$. (Why?)

OLS Asymptotics

- We know $E\left[X_i U_i\right] = 0$, so

$$\frac{1}{\sqrt{N}} \sum_i X_i U_i \xrightarrow{d} N\left(0, E\left[X_i X_i' U_i^2\right]\right)$$

- Therefore

$$\sqrt{N}\left(\hat{\beta} - \beta\right) \xrightarrow{d} N\left(0, E\left[X_i X_i'\right]^{-1} E\left[X_i X_i' U_i^2\right] E\left[X_i X_i'\right]^{-1}\right)$$

- This is the (first-order) asymptotic distribution of the OLS estimator with iid sampling.

Heteroskedasticity-robust std errors

$$\sqrt{N}\left(\hat{\beta}-\beta\right) \xrightarrow{d} N\left(0, E\left[X_i X_i'\right]^{-1} E\left[X_i X_i' U_i^2\right] E\left[X_i X_i'\right]^{-1}\right)$$

- Under homoskedasticity $E\left[U_i^2|X_i\right] = \sigma^2$ and variance collapses to $\sigma^2 E\left[X_i X_i'\right]^{-1}$
- Modern approach: avoid imposing assumptions on $E\left[U_i^2|X_i\right]$ and base inference on

$$\hat{V}\left(\hat{\beta}\right) = \frac{1}{N}\left(\frac{1}{N}\sum_i X_i X_i'\right)^{-1} \frac{1}{N}\sum_i X_i X_i' \hat{U}_i^2 \left(\frac{1}{N}\sum_i X_i X_i'\right)^{-1}$$

- This is the White (1980) heteroskedasticity-robust covariance matrix
- We assumed iid sampling, what if we relax this assumption?

Section 2

Clustering

Clustering SEs

- Traditionally assumed: observations are independent draws from a common distribution.
- Often implausible in serious economic research $\rightarrow$ dependent data (time series?!?).
- We now review how dependent data can affect inference of OLS (easy to generalize to other estimators, see ps)

Grouped Data

- Suppose the data have a group structure, with J groups and M_j individual observations in "cluster" j, for a total sample of $N = \sum_j M_j$.
- Regression model is

$$Y_{ij} = X_{ij}'\beta + U_{ij}$$

- We are worried that $Cov\left(U_{ij}, U_{i'j}\right) \neq 0$.

Clustering

- Recall that the OLS variance matrix is given by

$$V = (X'X)^{-1} X'\Omega X (X'X)^{-1}$$

- With correlation within but not between clusters of U_{ij} implies that the error variance matrix is block diagonal:

$$\mathrm{E}[UU'] = \Omega = \begin{bmatrix} \Omega_1 & 0 & 0 & 0 \\ 0 & \Omega_2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \Omega_J \end{bmatrix}$$

- Here Ω_j is the $M_j \times M_j$ covariance matrix for cluster j
- Can't get accurate estimate of each Ω_j. But to estimate the variance of OLS we only need the "meat" in the covariance sandwich:

$$X'\Omega X = \sum_j X_j' \Omega_j X_j$$

- Here X_j is a $M_j \times K$ matrix that stacks the Xs of all the observations belonging to cluster j.
- Obvious solution: treat the cluster level data vectors (X_j, Y_j) as primitives and apply usual White logic that we have covered before

Clustering

- Consistent estimate of the meat in the sandwich can be obtained by invoking a weak LLN for non-identically distributed data (Theorem 3.7 of White, 2001)

$$\frac{1}{N}\sum_j X_j'\hat{U}_j\hat{U}_j'X_j \xrightarrow{p} \lim_{J\to\infty}\frac{1}{N}\sum_j E\left[X_j'\Omega_j X_j\right]$$

- Recall that in OLS all randomess comes from the score $X'U = \sum_{j=1}^{J}\sum_{i=1}^{M_j} X_{ij}U_{ij}$.
- When normalized by $\frac{1}{N}$ the grand mean $\frac{1}{N}X'U$ is equivalent to a weighted average of cluster-specific means where the weights crucially depends on M_j.

Clustering

- Consistent estimate of the meat in the sandwich can be obtained by invoking a weak LLN for non-identically distributed data (Theorem 3.7 of White, 2001)

$$\frac{1}{N}\sum_j X_j'\hat{U}_j\hat{U}_j'X_j \xrightarrow{p} \lim_{J\to\infty}\frac{1}{N}\sum_j E\left[X_j'\Omega_j X_j\right]$$

- Recall that in OLS all randomess comes from the score $X'U = \sum_{j=1}^{J}\sum_{i=1}^{M_j} X_{ij}U_{ij}$.
- When normalized by $\frac{1}{N}$ the grand mean $\frac{1}{N}X'U$ is equivalent to a weighted average of cluster-specific means where the weights crucially depends on M_j.
- This generates the "cluster-robust" covariance matrix

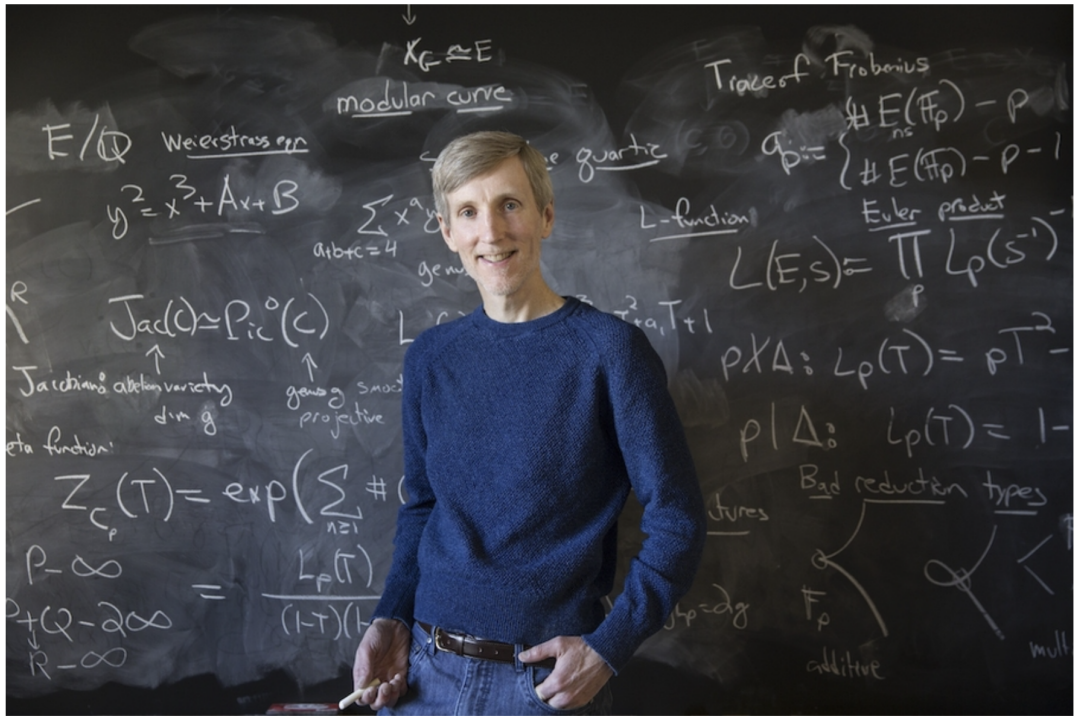
$$\hat{V}\left(\hat{\beta}\right) = \frac{1}{N}\left(\frac{1}{N}\sum_j X_j'X_j\right)^{-1}\left(\frac{1}{N}\sum_j X_j'\hat{U}_j\hat{U}_j'X_j\right)\left(\frac{1}{N}\sum_j X_j'X_j\right)^{-1}$$

- This approach relies on asymptotics in the number of groups; will perform poorly with few clusters
- Angrist and Pischke (2009): magic # of clusters is 42!

The answer to life, the universe, and everything

Mathematics researcher Drew Sutherland helps solve decades-old sum-of-three-cubes puzzle, with help from "The Hitchhiker's Guide to the Galaxy."

Sandi Miller | Department of Mathematics
September 10, 2019



A team led by Andrew Sutherland of MIT and Andrew Booker of Bristol University has solved the final piece of a famous 65-year old math puzzle with an answer for the most elusive number of all: 42.

The number 42 is especially significant to fans of science fiction novelist Douglas Adams' "The Hitchhiker's Guide to the Galaxy," because that number is the answer given by a supercomputer to "the Ultimate Question of Life, the Universe, and Everything."

Comments

- As usual, White SEs are biased in finite sample, this is why Stata implements a degrees of freedom correction to regression residuals

$$\tilde{U}_j = \sqrt{\frac{J}{J-1}\frac{N-1}{N-K}}\hat{U}_j \tag{1}$$

and replaces $\hat{U}_j$ with $\tilde{U}_j$ when computing $\hat{V}\left(\hat{\beta}\right)$.

Comments

- As usual, White SEs are biased in finite sample, this is why Stata implements a degrees of freedom correction to regression residuals

$$\tilde{U}_j = \sqrt{\frac{J}{J-1}\frac{N-1}{N-K}}\hat{U}_j \tag{1}$$

  and replaces $\hat{U}_j$ with $\tilde{U}_j$ when computing $\hat{V}\left(\hat{\beta}\right)$.

- Key question: at what level to cluster the standard error?
  - Important: being worried about $Cov\left(U_{ij}, U_{i'j}\right) \neq 0$ does not *necessarily* means that you have to cluster the standard errors.
  - What one should be worried about is whether $Cov\left(X_{ij}U_{ij}, X_{i'j}U_{i'j}\right) \neq 0$.

Comments

- As usual, White SEs are biased in finite sample, this is why Stata implements a degrees of freedom correction to regression residuals

$$\tilde{U}_j = \sqrt{\frac{J}{J-1}\frac{N-1}{N-K}}\hat{U}_j \tag{1}$$

  and replaces $\hat{U}_j$ with $\tilde{U}_j$ when computing $\hat{V}\left(\hat{\beta}\right)$.

- Key question: at what level to cluster the standard error?
  - Important: being worried about $Cov\left(U_{ij}, U_{i'j}\right) \neq 0$ does not *necessarily* means that you have to cluster the standard errors.
  - What one should be worried about is whether $Cov\left(X_{ij}U_{ij}, X_{i'j}U_{i'j}\right) \neq 0$.
- Rule of thumb: cluster at the level at which your treatment varies.

Example 1: Individual level RCT

- Example: you randomize cash transfers to individuals living in different villages in India.
- In this application: i indexes individuals, j indexes clusters.
- You want to run the following regression

$$Y_i = \alpha + \beta X_{ij} + e_{ij} \tag{2}$$

- Note that here if the treatment variable is randomized at the individual level $\rightarrow \mathrm{E}[X_{ij}X_{i'j}] = 0$ (suppose the X variable has been demeaned).
- This in turn implies that (by LIE) $\mathrm{E}[X_{ij}U_{ij}X_{i'j}U_{i'j}] = \mathrm{E}[X_{ij}X_{i'j}\,\mathrm{E}[U_{ij}U_{i'j}|X_{ij}X_{i'j}]] = 0 \rightarrow$ no need to cluster!

Example 2: Clustered RCT

- Example: you randomize cash transfers to different villages in India.
- You still have data on individuals as well as villages and are interested in running

$$Y_i = \alpha + \beta X_{ij} + e_{ij} \tag{3}$$

- Now $\mathrm{E}[X_{ij}X_{i'j}] \neq 0$ (treatment is perfectly correlated within clusters!).
- This in turn implies that $\mathrm{E}[X_{ij}U_{ij}X_{i'j}U_{i'j}] = \mathrm{E}[X_{ij}X_{i'j}\,\mathrm{E}[U_{ij}U_{i'j}|X_{ij}X_{i'j}]] \neq 0 \rightarrow$ now you have to cluster the SEs (at the village level)

Example 3: Real Life

- Most cases treatment of interest might vary within clusters.
- Example: you have data on individuals observed within each state across different years.
- Treatment is a policy implemented by a given state in a particular year (and remains active thereafter).
- Treatment therefore varies at the state-by-year level.
- However, since treatment is persistent, people often cluster at the level of the State, although such SEs might be too conservative (Bertrand, Duflo and Mullainathan, 2004; Abadie et al., 2023).

Key Reference

THE
QUARTERLY JOURNAL
OF ECONOMICS

Vol. 138 2023 Issue 1

WHEN SHOULD YOU ADJUST STANDARD ERRORS FOR CLUSTERING?*

ALBERTO ABADIE
SUSAN ATHEY
GUIDO W. IMBENS
JEFFREY M. WOOLDRIDGE

Clustered standard errors, with clusters defined by factors such as geography, are widespread in empirical research in economics and many other disciplines. Formally, clustered standard errors adjust for the correlations induced by sampling the outcome variable from a data-generating process with unobserved cluster-level components. However, the standard econometric framework for clustering leaves important questions unanswered: (i) Why do we adjust standard errors for clustering in some ways but not others, for example, by state but not by gender, and in observational studies but not in completely randomized experiments? (ii) Is the clustered variance estimator valid if we observe a large fraction of the clusters in the population? (iii) In what settings does the choice of whether and how to cluster make a difference? We address these and other questions using a novel framework for clustered inference on average treatment effects. In addition to the common sampling component, the new framework incorporates a design component that accounts for the variability induced on the estimator by the treatment assignment mechanism. We show that, when the number of clusters in the sample is a nonnegligible fraction of the number of clusters in the population, conventional clustered

Cluster SEs in Action

TABLE I

COLLEGE EFFECTS IN THE CENSUS SAMPLE

Panel A: Treatment: State indicator for share of some college greater than 0.55	OLS
Coefficient	0.1022
Standard error:	
Robust	(0.0012)
Cluster	(0.0312)

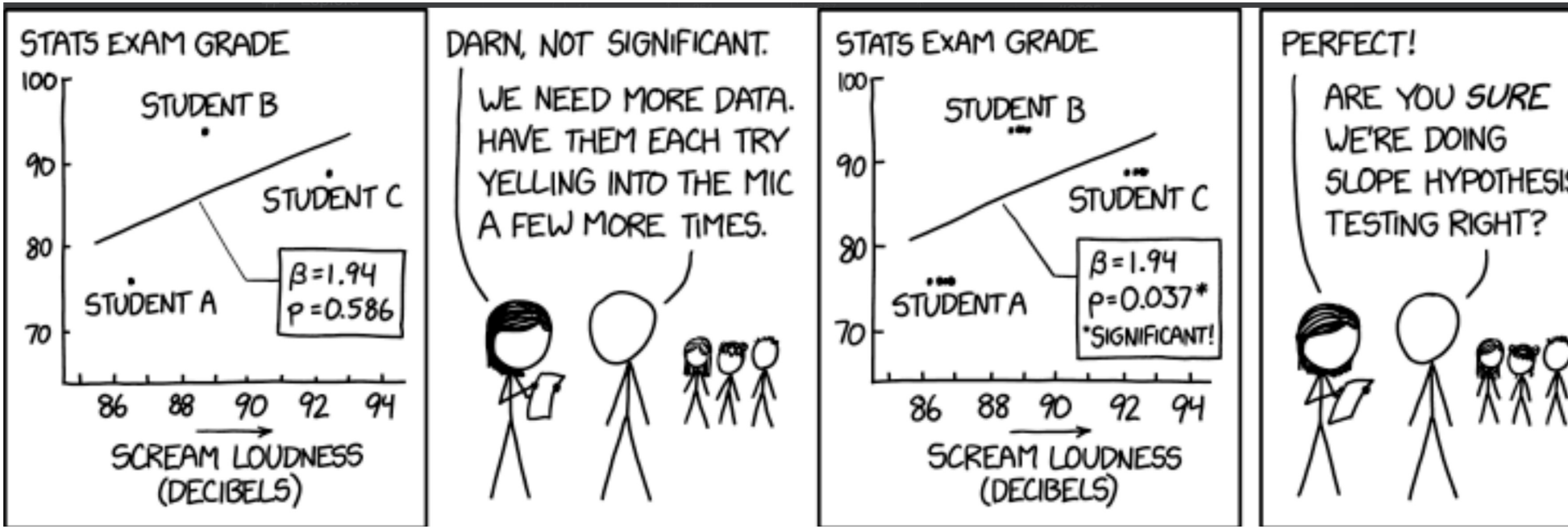
Panel B:	Treatment: Individual indicator for some college	
	OLS	FE
Coefficient	0.4656	0.4570
Standard error:		
Robust	(0.0012)	(0.0012)
Cluster	(0.0269)	(0.0276)
Causal cluster variance (CCV)	(0.0035)	(0.0014)
Two-stage cluster bootstrap (TSCB)	(0.0036)	(0.0014)

Notes. Dependent variable: log labor earnings. This table uses the 5 percent PUMS from the 2000 decennial census with Puerto Rico added. Log labor earnings are the log of annual earnings. The sample includes all individuals between 20 and 50 years of age with positive earnings. Some college is defined as 13 or more years of education. OLS refers to the least-squares estimator where the regressors include an intercept and the treatment indicator. FE refers to the fixed-effect estimator where the regression function also includes indicators for each of the clusters.

Takeaways from AAIW (2023)

- Figure it out at what level to cluster requires figuring out (i) the sampling process and (ii) how is treatment assigned.
- Thinking of a framework where $J \to \infty$ is often not useful $\to$ AAIW shows that is possible to derive standard errors smaller than those based on asymptotic approximation that refers to infinite super-population.
- Correlation in the regression error term within clusters does not automatically mean that one has to cluster (see example #1).
- The idea of "keep clustering at higher levels if SEs continue to become smaller" is ill-advised.
- Derive two new cluster-robust estimator of SEs that are considerably smaller than White-based cluster-robust SEs.
  - Especially valuable in contexts like Example 3 ("partially" clustered treatment assignment).

The need for clustering...



Section 3

Bootstrap

The Bootstrap



Preliminaries

- A test statistic can be written:

$$T_n \equiv T_n\left(\boldsymbol{z}_1, \boldsymbol{z}_2, ..., \boldsymbol{z}_N, F\right)$$

  where F is population distribution of data
- Example: "t-stat" is $T_n = \frac{\hat{\theta}-\theta_0}{\hat{\sigma}}$
- Finite sample dist of T_n is

$$G_n\left(t, F\right) \equiv P\left(T_n \leq t \middle| F\right)$$

- Problem: don't know F! (except under very strong null hypothesis)

Two approaches

Conventional asymptotics:

$$G_n(t, F) \approx \lim_{n \to \infty} G_n(t, F) = G(t) \quad (\text{"asymptotic pivot"})$$

- Example: "t-stat" goes to normal $G(t) = \Phi(t)$

Two approaches

Conventional asymptotics:

$$G_n(t, F) \approx \lim_{n\to\infty} G_n(t, F) = G(t) \quad (\text{"asymptotic pivot"})$$

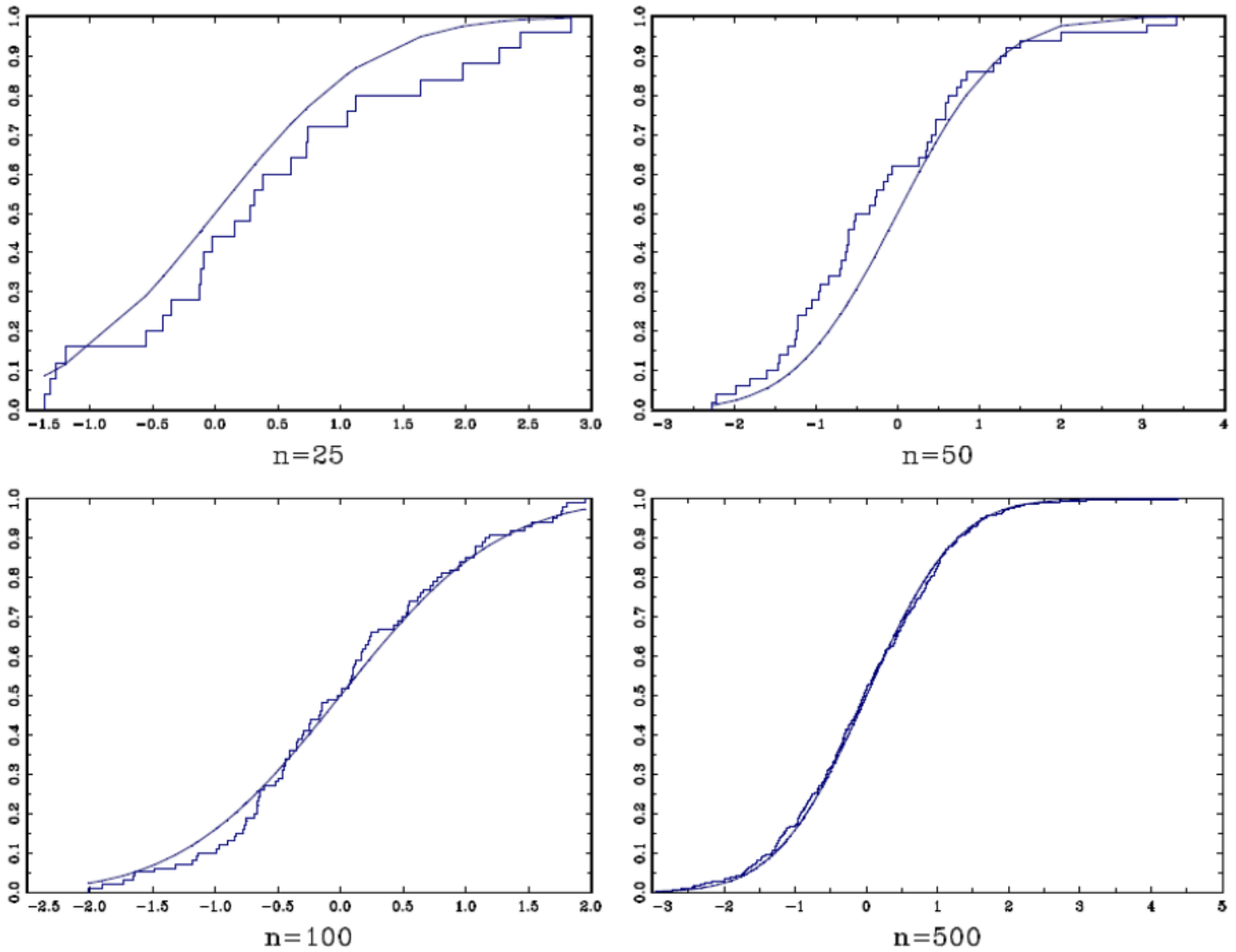
- Example: "t-stat" goes to normal $G(t) = \Phi(t)$

Bootstrap approximation:

$$G_n(t, F) \approx G_n(t, F_n)$$

- Traditional choice of F_n is EDF
- Glivenko-Cantelli: EDF converges uniformly to F

Convergence in action



Source: Hansen online textbook

Computation (Pairs Bootrstrap)

For small N, possible to enumerate $G_n(t, F_n)$ exactly (2^N atoms)
Otherwise use sampling w/ replacement to approximate $G_n(t, F_n)$

Computation (Pairs Bootrstrap)

For small N, possible to enumerate $G_n(t, F_n)$ exactly (2^N atoms)
Otherwise use sampling w/ replacement to approximate $G_n(t, F_n)$

Algorithm:

1. Draw a random sample of size N *with replacement* from the data. Call this the b'th "bootstrap" sample.
2. Compute the test statistic $T_n^{*(b)}$ in the bootstrap sample (see below for details on this).
3. Repeat #1 and #2 until $b = B$.

Computation (Pairs Bootrstrap)

For small N, possible to enumerate $G_n(t, F_n)$ exactly (2^N atoms)
Otherwise use sampling w/ replacement to approximate $G_n(t, F_n)$

Algorithm:

1. Draw a random sample of size N *with replacement* from the data. Call this the b'th "bootstrap" sample.
2. Compute the test statistic $T_n^{*(b)}$ in the bootstrap sample (see below for details on this).
3. Repeat #1 and #2 until $b = B$.

Some rules of thumb for B (from my personal experience):

- $B \geq 400$ for std error estimation
- $B \geq 999$ for quantiles

What if errors are serially correlated within a cluster?

Computation (Pairs Bootrstrap)

For small N, possible to enumerate $G_n(t, F_n)$ exactly (2^N atoms)
Otherwise use sampling w/ replacement to approximate $G_n(t, F_n)$

Algorithm:

1. Draw a random sample of size N *with replacement* from the data. Call this the b'th "bootstrap" sample.
2. Compute the test statistic $T_n^{*(b)}$ in the bootstrap sample (see below for details on this).
3. Repeat #1 and #2 until $b = B$.

Some rules of thumb for B (from my personal experience):

- $B \geq 400$ for std error estimation
- $B \geq 999$ for quantiles

What if errors are serially correlated within a cluster? Block Bootstrap.
What if I can't compute my model in a given bootrstrap replica?

Computation (Pairs Bootrstrap)

For small N, possible to enumerate $G_n(t, F_n)$ exactly (2^N atoms)
Otherwise use sampling w/ replacement to approximate $G_n(t, F_n)$

Algorithm:

1. Draw a random sample of size N *with replacement* from the data. Call this the b'th "bootstrap" sample.
2. Compute the test statistic $T_n^{*(b)}$ in the bootstrap sample (see below for details on this).
3. Repeat #1 and #2 until $b = B$.

Some rules of thumb for B (from my personal experience):

- $B \geq 400$ for std error estimation
- $B \geq 999$ for quantiles

What if errors are serially correlated within a cluster? Block Bootstrap.
What if I can't compute my model in a given bootrstrap replica? Residual/Wild Bootstrap.

When does the bootstrap work?

- Short answer: when things are "well behaved"
- Problems emerge when limiting distribution is not normal
- Example (Andrews, 2000):

$$\theta_0 = \max\{0, E[X_i]\}$$

$$T_n = \sqrt{N}\left(\max\left\{0, \overline{X}\right\} - \theta_0\right)$$

- T_n is not "regular" – limiting distribution is normal when $\theta_0 > 0$ but "half-normal" when $\theta_0 = 0$.
- Discontinuity at 0 creates problems for bootstrap
- Practical areas where this problem arises:
  - Partially identified models (Canay and Shaikh, 2016)
  - Nearest neighbor matching (Abadie and Imbens, 2008)

A useful guide..

Theorem (Fang and Santos, 2015)

If $\sqrt{N}\left(\hat{\theta}-\theta_0\right) \xrightarrow{d} N\left(0,\sigma\right)$ *and the bootstrap estimate* θ^* *obeys* $\sqrt{N}\left(\theta^*-\hat{\theta}\right) \xrightarrow{d} N\left(0,\sigma\right)$*, then the bootstrap is consistent for the limiting distribution of* $\sqrt{N}\left(f\left(\hat{\theta}\right)-f\left(\theta_0\right)\right)$ *iff* $f\left(.\right)$ *is differentiable.*

- "Delta method" for the bootstrap
- Rules out $\max\left\{.\right\}$ / $\min\left\{.\right\}$ transforms and "kinks"
- But allows smooth functions of sample moments.

When to use the bootstrap

- When computing an analytical standard error is tedious:
  - Quantile regression (avoid estimation of nuisance density)
  - Multi-step estimation (just bootstrap the entire procedure)

- When tests based on analytical standard error are poorly behaved (though other options are typically available):
  - Few clusters
  - Delta method for highly nonlinear hypothesis
  - Weak IV

Bootstrap refinements

- Bootstrapping t-stat often better than comparing t-stat to normal crit value (i.e. 1.96)
- Provides higher order approximation to $G_n(t, F)$ when $T_n(\theta) \equiv \frac{\widehat{\theta}-\theta}{\widehat{\sigma}}$ is an asymptotic pivot (Hall, 1992)
  - note: requires consistent std err estimator $\hat{\sigma}$!
- Algorithm:

1. Draw B bootstrap samples
2. Form BS test statistics $T_n^{*(b)}(\theta) \equiv \frac{\widehat{\theta}^{*(b)}-\theta}{\widehat{\sigma}^{*(b)}}$
3. Find $1-\alpha$'th quantile of $\left|T_n^{*(b)}\left(\hat{\theta}\right)\right|$, call it c^*
4. Reject if $|T_n(\theta_0)| > c^*$

Kline and Walters (2016, QJE)

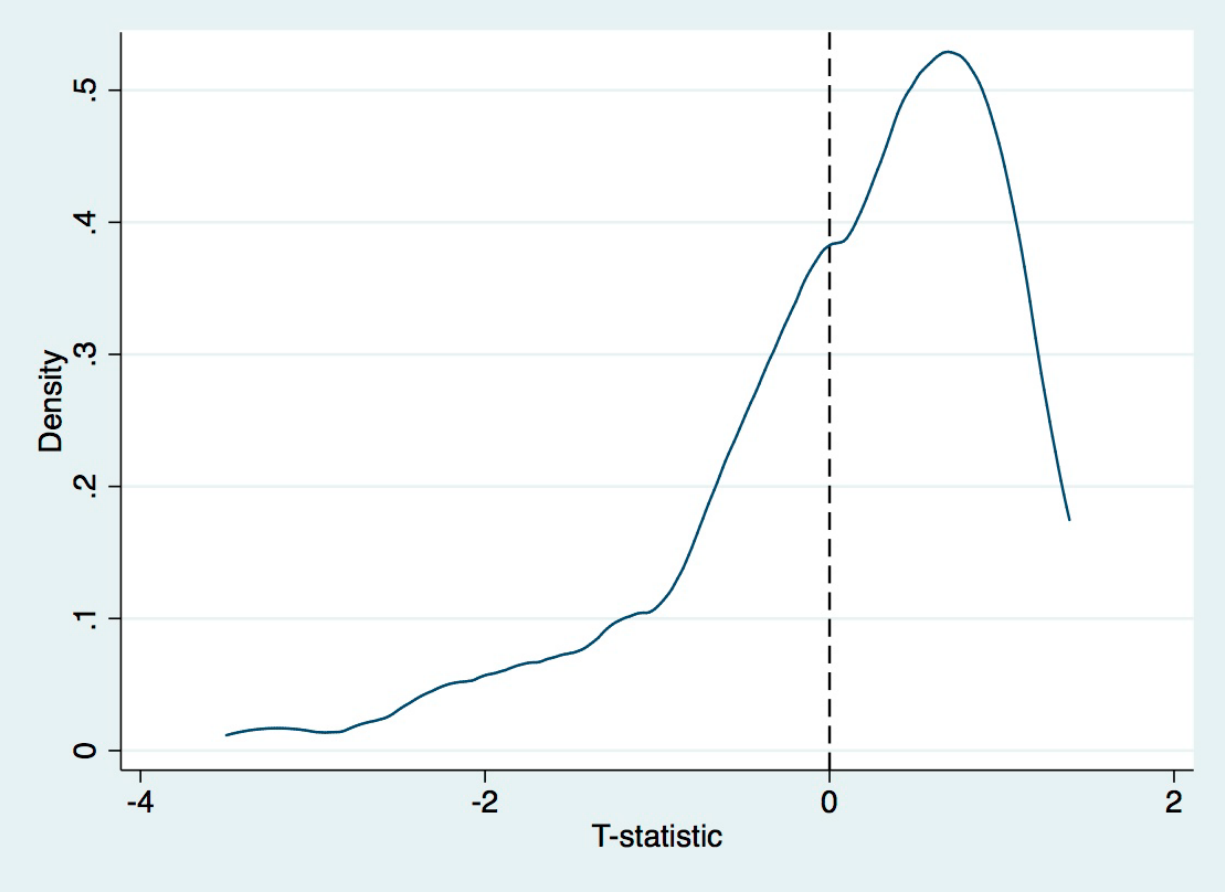
TABLE IV
(CONTINUED)

(1) Parameter	(2) Description	(3) Value	(4) Source
Panel B: Marginal value of public funds			
NMB	Marginal benefit to Head Start population net of taxes	\$5,513	$(1-\tau)pLATE_h$
MFC	Marginal fiscal cost of Head Start enrollment	\$5,031	$\varphi_h - \varphi_c S_c - \tau pLATE_h$, naive assumption
		\$3,671	Pessimistic assumption
		\$2,991	Preferred assumption
MVPF	Marginal value of public funds	1.10 (0.22)	*NMB/MFC* (std. err.), naive assumption
		p-value = .1	
			Breakeven $p/\overline{e} = 0.09\,(0.01)$
		1.50 (0.34)	Pessimistic assumption
		p-value = .00	
			Breakeven $p/\overline{e} = 0.08\,(0.01)$
		1.84 (0.47)	Preferred assumption
		p-value = .00	
			Breakeven $p/\overline{e} = 0.07\,(0.01)$

Notes. This table reports results of cost-benefit calculations for Head Start. Parameter values are obtained from the sources listed in column (4). Standard errors for MVPF ratios are calculated using the delta method. *P*-values are from one-tailed tests of the null hypotheses that the MVPF is less than 1. These tests are performed via nonparametric block bootstrap of the *t*-statistic, clustered at the Head Start center level. Breakevens give percentage effects of a standard deviation of test scores on earnings that set *MVPF* equal to 1.

- t-stat for $MVPF = 1$ in "pessimistic" scenario is $\frac{.5}{.34} = 1.47$
- Normal p-val for 1-sided test would be $1 - \Phi(1.47) = .07$. Refined p-val$\approx$ 0!

A monte carlo study



- Delta method normal approximation is poor here
  - Overcoverage: delta method rejected in 0/500 simulations
- Bootstrap picks up asymmetry in finite sample dist
  - Good coverage: 7% rejections on nominal 5% test

Residual Bootstraps

Residual bootstrap for linear model (Freedman, 1981)
Algorithm:

1. Fit linear model $Y_i = X_i\beta + \varepsilon_i$
2. Resample from residuals $\hat{\varepsilon}_i$ to make new dependent variable $Y_i^* = X_i\hat{\beta} + \varepsilon_i^*$
3. Fit linear model to Y_i^*, get β^*

Residual Bootstraps

Residual bootstrap for linear model (Freedman, 1981)
Algorithm:

1. Fit linear model $Y_i = X_i\beta + \varepsilon_i$
2. Resample from residuals $\hat{\varepsilon}_i$ to make new dependent variable $Y_i^* = X_i\hat{\beta} + \varepsilon_i^*$
3. Fit linear model to Y_i^*, get β^*

- Very good asymptotic properties
  - Can even allow $dim(\beta)$ to grow w/ N (Bickel and Freedman, 1983)
  - Intuition: resampling from a scalar instead of from (Y_i, X_i)
- Do you see a problem with this?

Wild Bootstrap (Wu, 1986; Liu, 1988):

Allow heteroscedasticity while retaining advantage of residual BS
Algorithm:

1. Fit linear model $Y_i = X_i\beta + \varepsilon_i$
2. Draw iid weights $\{W_i^*\}_{i=1}^N$ obeying $E[W_i^*] = 0$, $E\left[(W_i^*)^2\right] = 1$
3. Make new dependent variable $Y_i^* = X_i\hat{\beta} + \hat{\varepsilon}_i W_i^*$
4. Fit linear model to Y_i^*, get (β^*, σ^*)

Wild Bootstrap (Wu, 1986; Liu, 1988):

Allow heteroscedasticity while retaining advantage of residual BS
Algorithm:

1. Fit linear model $Y_i = X_i\beta + \varepsilon_i$
2. Draw iid weights $\{W_i^*\}_{i=1}^N$ obeying $E[W_i^*] = 0$, $E\left[(W_i^*)^2\right] = 1$
3. Make new dependent variable $Y_i^* = X_i\hat{\beta} + \hat{\varepsilon}_i W_i^*$
4. Fit linear model to Y_i^*, get (β^*, σ^*)

Intuition:

$$E\left[\hat{\varepsilon}_i W_i^* \,|\, \{Y_i, X_i\}_{i=1}^N\right] = 0$$

$$E\left[(\hat{\varepsilon}_i W_i^*)^2 \,|\, \{Y_i, X_i\}_{i=1}^N\right] = \hat{\varepsilon}_i^2$$

So $\varepsilon_i^* \equiv \hat{\varepsilon}_i W_i^*$ is mean independent of X_i and replicates hetero found in data
But do we want to impose mean independence?

Theoretical results

- Mammen (1993):
  - Wild BS yields refinement even under many regressors asymptotics ($p \to \infty,\ p/n \to 0$) provided linear model is correct
  - Wild (& "pairs") BS still consistent under misspecification
- Kline and Santos (2012, JoE): Wild BS retains refinement under fixed p asymptotics when "locally" misspecified
- Kline and Santos (2012, JEMS): "Score BS"
  - extension of Wild BS to nonlinear models (e.g. Probit)
  - avoid recomputation of estimator (major speedup)
  - connection to Bayesian/Weighted BS
  - implemented in "BOOTTEST" Stata package

Cameron, Gelbach, and Miller (2008, RESTAT)

- Monte Carlo bootstrap tests in clustered data setting
- Variant of Wild BS that "imposes the null" performs best

Algorithm:

1. Fit restricted linear model $Y_i = X_i\tilde{\beta} + \varepsilon_i$ where $c\left(\tilde{\beta}\right) = 0$
2. Draw iid weights $\{W_i^*\}_{i=1}^N$ obeying $E\left[W_i^*\right] = 0$, $E\left[\left(W_i^*\right)^2\right] = 1$
3. Make new dependent variable $Y_i^* = X_i\hat{\tilde{\beta}} + \hat{\tilde{\varepsilon}}_i W_i^*$
4. Fit unrestricted linear model to Y_i^*, get $\left(\beta^*, \sigma^*\right)$
5. Compare clustered Wald stat $\hat{T}_n$ to appropriate quantile of bootstrap analogue T_n^*

Cameron, Gelbach, and Miller (2008, RESTAT)

- Monte Carlo bootstrap tests in clustered data setting
- Variant of Wild BS that "imposes the null" performs best

Algorithm:

1. Fit restricted linear model $Y_i = X_i\tilde{\beta} + \varepsilon_i$ where $c\left(\tilde{\beta}\right) = 0$
2. Draw iid weights $\{W_i^*\}_{i=1}^N$ obeying $E\left[W_i^*\right] = 0$, $E\left[\left(W_i^*\right)^2\right] = 1$
3. Make new dependent variable $Y_i^* = X_i\hat{\tilde{\beta}} + \hat{\tilde{\varepsilon}}_i W_i^*$
4. Fit unrestricted linear model to Y_i^*, get $\left(\beta^*, \sigma^*\right)$
5. Compare clustered Wald stat $\hat{T}_n$ to appropriate quantile of bootstrap analogue T_n^*

- Canay, Shaikh, Santos (2016): CGM BS consistent for fixed # of clusters as cluster size grows, given balanced design ($E\left[X_i X_i' | C_i\right] = E\left[X_i X_i'\right]$) and effect homogeneity $\beta_c = \beta$.

Beware!

Can We Trust the Bootstrap in High-dimensions? The Case of Linear Models

Noureddine El Karoui NKAROUI@BERKELEY.EDU, N.ELKAROUI@CRITEO.COM
Criteo AI Lab
32 Rue Blanche
75009 Paris, France
and
Department of Statistics
University of California
Berkeley, CA 94270, USA

Elizabeth Purdom EPURDOM@STAT.BERKELEY.EDU
Department of Statistics
University of California
Berkeley, CA 94270, USA

Editor: Guy Lebanon

(The answer is no)

Can We Trust the Bootstrap in High-dimensions? The Case of Linear Models

Noureddine El Karoui NKAROUI@BERKELEY.EDU, N.ELKAROUI@CRITEO.COM
Criteo AI Lab
32 Rue Blanche
75009 Paris, France
and
Department of Statistics
University of California
Berkeley, CA 94270, USA

Elizabeth Purdom EPURDOM@STAT.BERKELEY.EDU
Department of Statistics
University of California
Berkeley, CA 94270, USA

Editor: Guy Lebanon

Section 4

Two-Step Estimators

2-step estimators

Lots of settings where it is easier to estimate model in "steps"

- 1st step parameter estimate $\hat{\gamma}$
- Second step m-estimator:

$$\widehat{\theta} \equiv \arg\min_{\theta \in \Theta} \sum_i q\left(\boldsymbol{Z}_i, \theta; \widehat{\gamma}\right)$$

Example (WNLLS):

$$\widehat{\theta} = \arg\min_{\theta \in \Theta} \sum_i \left[Y_i - m\left(X_i, \theta\right)\right]^2 w\left(X_i; \widehat{\gamma}\right)$$

- $\hat{\gamma} \xrightarrow{p} \gamma^*$
- How/when to adjust standard errors for use of $\hat{\gamma}$ instead of γ^*?

Influence fn representation

Asymptotic variability derives from two sources:

$$\begin{aligned}\sqrt{N}\left(\widehat{\theta}-\theta_0\right) &= -E\left[\boldsymbol{H}_i\left(\theta_0,\gamma^*\right)\right]^{-1} N^{-1/2}\sum_i \boldsymbol{s}_i\left(\theta_0,\gamma^*\right) \\ &\quad -E\left[\boldsymbol{H}_i\left(\theta_0,\gamma^*\right)\right]^{-1} E\left[\bigtriangledown_\gamma \boldsymbol{s}_i\left(\theta_0,\gamma^*\right)\right]\sqrt{N}\left(\widehat{\gamma}-\gamma^*\right) \\ &\quad +o_p\left(1\right)\end{aligned}$$

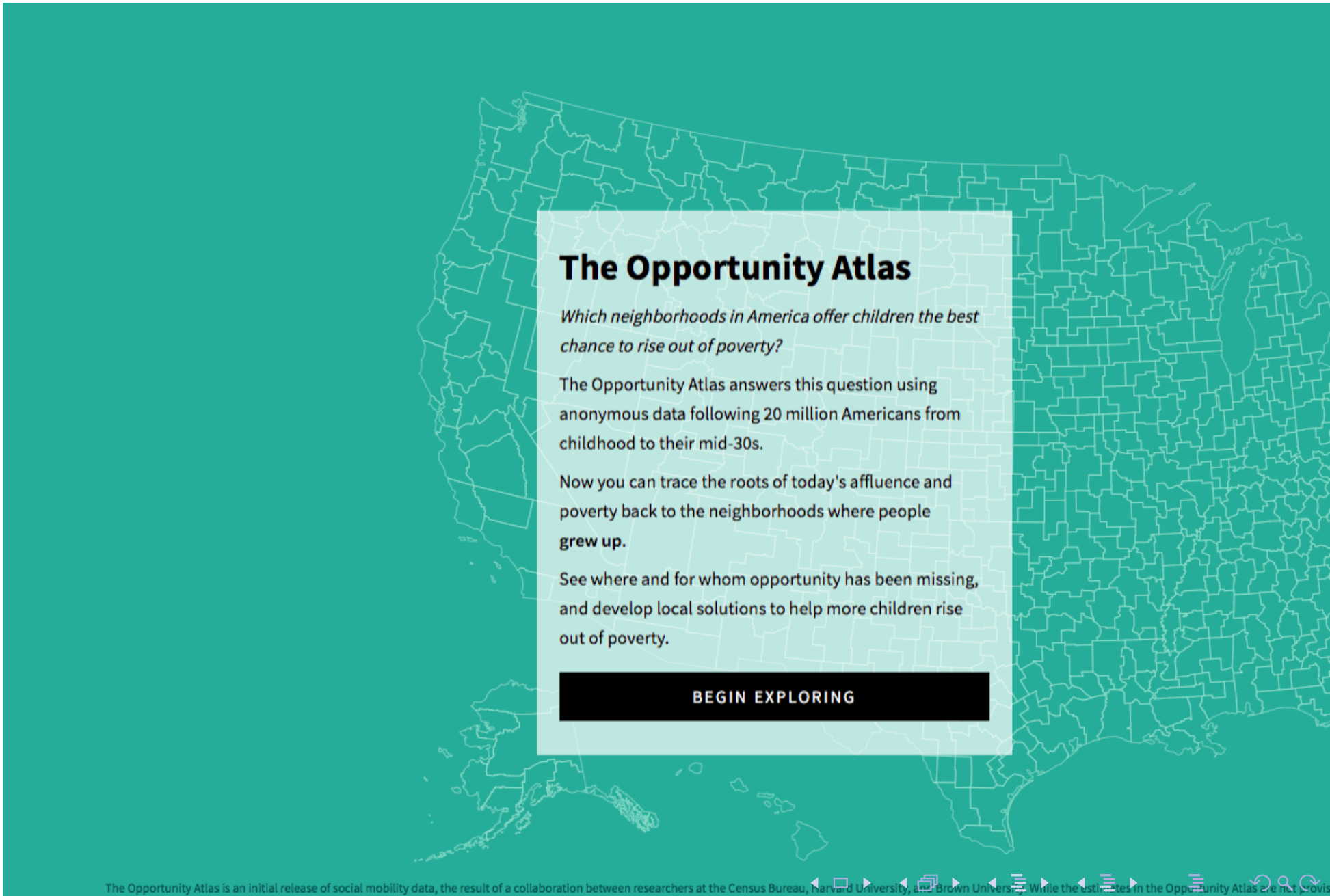
Second term is zero when $E\left[\bigtriangledown_\gamma \boldsymbol{s}_i\left(\theta_0,\gamma^*\right)\right]=0$!

Example (WNLLS):

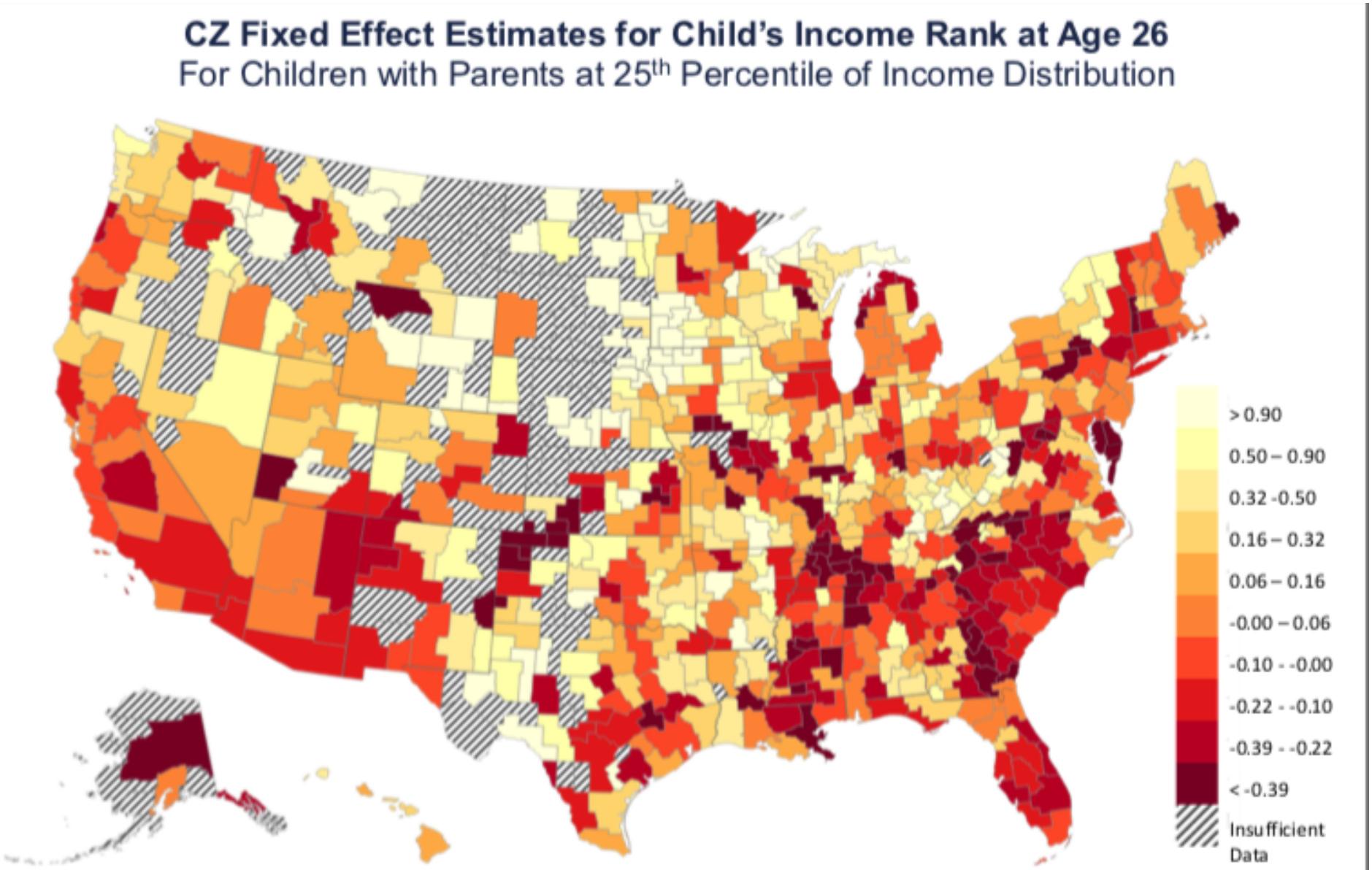
$$\bigtriangledown_\gamma \boldsymbol{s}_i\left(\theta_0,\gamma^*\right)=\left[Y_i-m\left(X_i,\theta_0\right)\right]\frac{\partial}{\partial\theta}m\left(X_i,\theta_0\right)\bigtriangledown_\gamma w\left(X_i;\gamma^*\right)$$

- If $E\left[Y_i|X_i\right]=m\left(x_i,\theta_0\right)$ then $E\left[\bigtriangledown_\gamma \boldsymbol{s}_i\left(\theta_0,\gamma^*\right)\right]=0$ (by LIE)
- Otherwise, all bets are off

Two-Steps estimators are everywhere



Place "effects"



Two-step in action

ONLINE APPENDIX TABLE XII

Associations Between Place Effects and Area-Level Characteristics Across Counties within CZs for Low-Income Families (p = 25)

		SD of Covariate (1)	Signal Correlation (2)		Regression Estimates					
					Perm. Res. (3)		Causal (4)		Sorting (5)	
		Std. Dev	Correlation	s.e.	%	s.e.	%	s.e.	%	s.e.
Segregation and Poverty	Fraction Black Residents	0.130	-0.32	0.10	-7.07	0.55	-1.98	0.64	-5.08	0.69
	Poverty Rate	0.055	-0.23	0.11	-6.12	0.72	-1.44	0.67	-4.67	0.63
	Racial Segregation Theil Index	0.118	-0.37	0.10	-7.01	0.46	-2.31	0.60	-4.70	0.61
	Income Segregation Theil Index	0.039	-0.42	0.10	-5.25	0.36	-2.62	0.63	-2.63	0.62
	Segregation of Poverty (<p25)	0.034	-0.46	0.10	-5.65	0.41	-2.88	0.64	-2.77	0.65
	Segregation of Affluence (>p75)	0.045	-0.36	0.11	-4.53	0.39	-2.22	0.66	-2.31	0.62
	Share with Commute < 15 Mins	0.100	0.02	0.12	0.73	0.62	0.12	0.73	0.62	0.98
	Log. Population Density	1.698	-0.27	0.11	-5.53	0.86	-1.67	0.69	-3.86	0.93
Income Distribution	Household Income per Capita for Working-Age Adults	9220.2	0.06	0.14	2.56	0.79	0.35	0.87	2.20	0.62
	Gini coefficient for Parent Income	0.113	-0.41	0.14	-6.05	1.31	-2.55	0.85	-3.50	0.86
	Top 1% Income Share for Parents	0.001	-0.23	0.10	-2.96	0.82	-1.41	0.59	-1.54	0.74
	Gini within Bottom 99%	0.112	-0.41	0.14	-6.06	1.31	-2.55	0.85	-3.51	0.86
	Fraction Middle Class (Between National p25 and p75)	0.074	0.13	0.13	2.14	0.83	0.80	0.83	1.34	0.72
Tax	Local Tax Rate	0.009	-0.21	0.12	-2.82	2.03	-1.32	0.77	-1.51	1.71
	Local Tax Rate per Capita	0.432	-0.15	0.11	-1.34	1.63	-0.91	0.66	-0.44	1.47
	Local Government Expenditures per Capita	1.0	-0.30	0.13	-3.26	1.80	-1.86	0.84	-1.40	1.37
	State EITC Exposure	3.747	-0.01	0.21	-0.27	0.19	-0.08	1.31	-0.19	1.23
	State Income Tax Progressivity	2.368	-0.19	0.27	-0.42	0.41	-1.20	1.68	0.78	1.80
K-12 Education	School Expenditure per Student ($1000)	1.478	-0.07	0.12	-1.13	1.29	-0.41	0.75	-0.73	1.23
	Student/Teacher Ratio	2.807	-0.10	0.11	-1.73	0.71	-0.65	0.67	-1.08	0.94
	Test Score Percentile (Controlling for Parent Income)	9.593	0.35	0.13	5.51	1.16	2.20	0.81	3.31	0.99
	High School Dropout Rate (Controlling for Parent Income)	0.023	-0.37	0.13	-5.63	0.69	-2.33	0.80	-3.30	0.95
College	Number of Colleges per Capita	0.011	-0.19	0.16	-3.10	0.42	-1.15	1.01	-1.95	0.97
	Mean College Tuition	4421.3	-0.02	0.14	-1.03	0.79	-0.10	0.86	-0.93	1.24
	College Graduation Rate (Controlling for Parent Income)	0.139	0.03	0.16	-1.71	0.63	0.22	0.97	-1.93	1.06